An NLP-Driven System for Semantic Classification and Personalized Study Plan Generation

Project ID: 25-26J-089

Project Proposal Report

Kajamalan S – IT22113672

B.Sc. (Hons) Degree in Information Technology Specializing in Data Science

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

AUGUST-2025

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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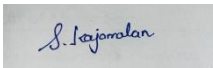
Kajamalan S	IT22113672		19/09/2025

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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(Dr. Sanvitha Kasthuriarachchii) Date----------

Signature of the Co-supervisor:

(Ms. Karthiga Rajendran) Date----------

ABSTRACT

This research project proposes an AI-powered educational framework utilizing Natural Language Processing (NLP) and educational analytics to optimize curriculum alignment and exam preparation for the Database Management Systems (DMS) module at the Sri Lanka Institute of Information Technology (SLIIT). The proposed system addresses significant challenges commonly faced by DMS students, such as aligning unstructured lecture materials with hierarchical syllabus modules, identifying recurring patterns across content, and prioritizing high-impact topics for more effective and efficient exam preparation. By analysing a decade of SLIIT DMS exam papers alongside lecture content, the framework leverages Sentence-BERT for semantic classification, enabling accurate mapping of questions and instructional materials to the most relevant syllabus modules. Frequency analysis is applied to detect recurring, high-priority topics, thereby guiding study prioritization in a structured and systematic manner. A scheduling algorithm dynamically generates personalized, time-aware study plans based on student-provided inputs, including exam dates, and daily study hours. Furthermore, a graph-based similarity search powered by GraphRAG recommends highly relevant practice questions per topic, which are tagged by difficulty level and syllabus module to support more targeted, adaptive, and effective revision strategies. Delivered through an interactive Streamlit interface, the system integrates multiple features such as countdown timers, topic priority visualizations, progress trackers, and intuitive search and filtering options, all of which enhance student engagement, motivation, usability, and overall learning experience. Designed with scalability and sustainability in mind, it processes large datasets efficiently using Dask, while ensuring strict data privacy and compliance through local processing, thus minimizing risks associated with external storage. The modular architecture allows seamless adaptation and extension to other SLIIT modules, addressing curriculum misalignment across different subjects and ultimately improving student outcomes. Rigorous validation ensures consistently high classification accuracy, positive user satisfaction, and robust performance across diverse datasets, resulting in a privacy-compliant, scalable, and sustainable solution that significantly enhances exam preparation efficiency, long-term knowledge retention, and measurable academic performance for DMS students.

Keywords: Natural Language Processing, Semantic Classification, Personalized Study Plans, Sentence-BERT, GraphRAG, Educational Analytics, Recommends practice questions, Database Management Systems, Curriculum Alignment, Streamlit, Data Privacy, Exam Preparation, SLIIT, Machine Learning, Frequency Analysis

TABLE OF CONTENTS

DECLARATION ..... - 3 -

ABSTRACT ..... - 4 -

TABLE OF CONTENTS…… … -5 -

List of Figures ..... - 6 -

1.0 INTRODUCTION ..... - 7 -

1.1 Background & Literature Survey ..... - 8 -

1.2 Research Gap ..... - 9 -

1.3 Research Problem ..... - 11 -

2.0 OBJECTIVES ..... - 12 -

2.1 Main Objectives ..... - 12 -

2.2 Specific Objectives ..... - 12 -

3.0 METHODOLOGY ..... - 14 -

3.1 System description ..... - 14 -

3.2 Key Components ..... - 14 -

3.3 System Workflow ..... - 16 -

3.4 Tools & Technology ..... - 17 -

Overall System Diagram ..... - 18 -

Component Diagram …… - 19 -

Work breakdown Structure…… - 20 -

3.4 PROJECT REQUIREMENTS ..... - 21 -

3.4.1 Functional requirements ..... - 21 -

3.4.2 Non-functional requirements ..... - 22 -

3.5 Expected Test Cases…… - 23 -

3.6 Timeline…… - 24 -

3.7 Risk Management Plan……- 25 -

4.0 COMMERCIALIZATION…… …….. - 26 -

5.0 BUDGET & BUDGET JUSTIFICATION ..... - 27 -

6.0 SUMMARY…… …… - 28 -

7.0 References ..... - 29 -

List of Figures

Figure 1- Competitive Analysis .....-10-**Error! Bookmark not defined.**
Figure 2- Study Plan System ..... - 10 -
Figure 3: System Diagram ..... **Error! Bookmark not defined.**
Figure 4: Component Diagram ..... - 20 -
Figure 4: Work breakdown Structure..... - 20 -
Figure 5:Gantt Chart ..... - 24 -

1.0 INTRODUCTION

Assessment is one of the most influential drivers of student learning in higher education, yet many universities struggle to maintain consistent alignment between curriculum delivery, lecture content, and examination practices. This misalignment often results in students devoting significant time to topics that appear only rarely in exams, while overlooking those that are frequently tested and more critical for success. As a consequence, students' preparation becomes inefficient and learning outcomes are not fully achieved. Traditional approaches, such as manually analyzing past exam papers and lecture materials, are often subjective, inconsistent, and time-consuming, making them inadequate for addressing this issue on a larger scale.

The emergence of Artificial Intelligence (AI), Natural Language Processing (NLP), and educational analytics presents a timely opportunity to transform this process. Unlike conventional learning management systems, which largely focus on content delivery and administrative functions, AI-powered approaches can uncover structural and topical trends across years of academic assessments, identify gaps between taught material and examined content, and provide actionable insights for both learners and educators. These capabilities not only optimize revision strategies for students but also guide institutions in improving curriculum design and instructional effectiveness.

Our research targets the development of an intelligent, AI-powered academic support framework that systematically analyzes historical exam papers, lecture content, and continuous assessments to deliver data-driven insights and personalized learning outputs. Through advanced NLP techniques, the system identifies high-frequency topics, predicts question difficulty, and classifies content according to syllabus requirements, while also highlighting redundancies and underrepresented areas. Based on these analyses, the framework generates practical learning aids, including model exam papers that reflect historical patterns, concise and curriculum-aligned short notes, adaptive essay evaluation with feedback, interactive flashcards, voice notes, and personalized time-aware study schedules.

By integrating these elements into a unified platform, the research provides a holistic solution that enhances student self-learning, improves exam preparation efficiency, and strengthens curriculum alignment with assessment practices. Ultimately, the project seeks to bridge the gap between academic delivery and evaluation through a data-driven, scalable, and personalized educational support system, thereby contributing to improved student achievement and institutional teaching quality.

1.1 Background & Literature Survey

The integration of Artificial Intelligence (AI) and Natural Language Processing (NLP) into higher education offers transformative potential for aligning curriculum delivery with examination demands, particularly for technical modules like Database Management Systems (DMS) at institutions such as the Sri Lanka Institute of Information Technology (SLIIT). Students frequently struggle to align unstructured lecture notes with hierarchical syllabus structures and prioritize high-impact exam topics, often relying on manual, time-consuming methods that overlook historical exam trends and individual time constraints. Existing research highlights AI's educational applications but reveals gaps in exam-specific solutions. For instance, Vo et al. (2022) developed CSIT-NER to align curricula with industry needs via NLP, yet it lacks exam question classification [1]. Chu and Ashraf (2025) applied predictive analytics for student retention, missing semantic classification or exam-focused scheduling [2]. Wang and Yu (2021) explored data-driven course sequencing, but their work omits NLP or exam-specific personalization [3]. These studies underscore AI's potential but fail to address tailored DMS exam preparation through semantic mapping or personalized study planning.

To address this challenge, mastery of key techniques and foundational studies is critical. Sentence-BERT, as demonstrated by Reimers and Gurevych (2019), is essential for generating semantic embeddings to map exam questions and lecture content to DMS syllabus modules [6]. Fine-tuning classifiers like BERT, as shown by Devlin et al. (2018), ensures accurate syllabus alignment [9]. TF-IDF-based frequency analysis, applied in educational data mining by Romero and Ventura (2020), enables identification of high-priority topics from historical exam data [8]. Scheduling algorithms, inspired by operations research, are necessary for creating time-aware study plans tailored to student availability. The GraphRAG library, a latest advancement from Microsoft (2024), supports efficient graph-based similarity searches for recommending practice questions by leveraging knowledge graphs to capture complex relationships. Additionally, proficiency in Streamlit for building interactive interfaces and Dask for scalable data processing, as used in large-scale educational analytics, is vital. These techniques, grounded in prior research, ensure a robust system tailored to SLIIT's DMS module.

Current state-of-the-art AI-driven educational tools focus on adaptive learning and curriculum alignment but lack exam-specific personalization for technical subjects like DMS. Sajja et al. (2024) introduced an AI assistant for quiz generation, yet it neglects hierarchical syllabus mapping or historical exam analysis [4]. Backenköhler et al. (2018) proposed personalized curricula through course sequencing, omitting NLP-driven classification or exam-focused strategies [5]. In contrast, the proposed system advances these efforts by integrating Sentence-BERT for semantic classification, analyzing a decade of SLIIT DMS exam data for frequency trends, and employing scheduling algorithms for personalized study plans. It distinguishes itself with a Streamlit-based interface featuring interactive elements like countdown timers and topic priority visualizations, enhancing student engagement. By incorporating the latest technology like GraphRAG for graph-based question recommendation, the system processes data locally to ensure privacy and adopts a modular design for scalability, addressing curriculum-examination misalignment, optimizing exam preparation efficiency, and delivering a novel, targeted solution for SLIIT DMS students, building on prior research with practical, exam-centric innovations.

1.2 Research Gap

The integration of Artificial Intelligence (AI) and Natural Language Processing (NLP) into higher education has revolutionized curriculum development and individualized learning, yet notable deficiencies remain in tackling exam-focused preparation for technical subjects such as Database Management Systems (DMS) at the Sri Lanka Institute of Information Technology (SLIIT).Current research often emphasizes broad curriculum alignment or industry-focused skill mapping, overlooking the specific needs of DMS students to align unstructured lecture content with hierarchical syllabus structures, analyze historical exam patterns, and create time-sensitive study plans tailored to exam deadlines. These deficiencies lead to inefficient study strategies, as students struggle to prioritize high-priority topics and align their preparation with exam requirements, resulting in suboptimal academic outcomes. The lack of integrated, exam-centric tools designed for SLIIT's DMS module underscores a critical gap in the literature, particularly in addressing the challenges of technical exam preparation in resource-limited academic environments.

Review of existing studies highlights these limitations. Vo et al. (2022) introduced CSIT-NER for extracting skills from course descriptions to align curricula with industry demands, but it lacks mechanisms for exam question classification or historical trend analysis [1]. Chu and Ashraf (2025) employed predictive analytics to enhance student retention, yet their framework omits semantic classification or exam-focused scheduling [2]. Wang and Yu (2021) proposed data-driven course sequencing for online education, without incorporating NLP techniques or exam-specific applications [3]. Sajja et al. (2024) developed an AI assistant for adaptive quiz generation, but it does not support hierarchical syllabus mapping or historical exam analysis [4]. Backenköhler et al. (2018) focused on personalized curricula via course sequencing, neglecting NLP-driven classification or exam-tailored strategies [5]. These studies collectively fail to provide an integrated solution combining semantic content mapping, historical exam analysis, and personalized study planning for DMS exam preparation at SLIIT.

This research addresses these gaps by developing an NLP-driven framework specifically for SLIIT's DMS module, delivering a targeted solution to enhance exam preparation efficiency and curriculum alignment. The proposed system utilizes Sentence-BERT for semantic classification to map exam questions and lecture materials to syllabus modules, applies TF-IDF-based frequency analysis on a decade of SLIIT DMS exam data to identify high-impact topics, and implements a scheduling algorithm to generate personalized, time-aware study plans based on student inputs such as exam dates and available study hours. A graph-based similarity search, powered by GraphRAG, recommends practice questions tagged by difficulty and module. Delivered through a Streamlit-based interface with interactive features like countdown timers and topic priority visualizations, the system ensures scalability, privacy via local data processing, and

adaptability to other SLIIT modules. By integrating exam-specific personalization and historical trend analysis, this framework offers a scalable, user-centric solution to improve academic performance and engagement for DMS students.

Feature	Research 01 [1]	Research 02 [2]	Research 03 [3]	Research 04 [4]	Research 05 [5]	Proposed System
Semantic Classification of Exam Content	✗	✗	✗	✗	✗	✓
Hierarchical Syllabus Mapping	✗	✗	✗	✗	✓	✓
Historical Exam Trend Analysis	✗	✗	✗	✗	✗	✓
Time-Aware Study Plans	✗	✗	✗	✗	✓✓	✓
Practice Question Recommendation	✗	✗	✓	✓	✗	✓
Exam-Focused Preparation	✗	✗	✗	✗	✗	✓

Figure 1- Competitive Analysis

1.3 Research Problem

Students in the Database Management Systems (DMS) module at the Sri Lanka Institute of Information Technology (SLIIT) encounter significant obstacles in exam preparation, including aligning unstructured lecture materials with hierarchical syllabus modules and identifying high-priority topics based on historical exam trends. Manual approaches to these tasks are labour-intensive, prone to errors, and fail to consider individual time constraints, such as exam dates and available daily study hours, resulting in inefficient revision strategies and reduced academic performance. The absence of automated tools tailored specifically for SLIIT's DMS module exacerbates these challenges, forcing students to rely on generic study methods that do not leverage historic exam data or provide personalized scheduling, hindering effective preparation and curriculum alignment.

Current AI-driven educational systems fall short in addressing these issues. Vo et al. (2022) developed CSIT-NER for skill extraction from course descriptions, aligning curricula with industry needs but lacking mechanisms for classifying exam questions or analysing historical trends [1]. Chu and Ashraf (2025) utilized predictive analytics to improve student retention, yet their system does not support semantic classification or time-aware exam scheduling [2]. Wang and Yu (2021) focused on course sequencing for online delivery, omitting NLP techniques or exam-specific applications [3]. Sajja et al. (2024) proposed an AI assistant for adaptive quiz generation, but it lacks hierarchical syllabus mapping or historical exam analysis [4]. Backe Köhler et al. (2018) explored personalized curricula through course sequencing, without incorporating NLP-driven classification or exam-focused personalization [5]. These systems collectively fail to provide an integrated solution for the exam preparation needs of SLIIT's DMS module.

This research addresses these gaps by developing an NLP-driven framework tailored for SLIIT's DMS module to optimize exam preparation efficiency and curriculum alignment. Leveraging Sentence-BERT for semantic classification, it maps exam questions and lecture materials to syllabus modules. TF-IDF-based frequency analysis of a decade of SLIIT DMS exam data identifies high-impact topics, while a scheduling algorithm generates personalized, time-aware study plans based on student inputs, such as exam dates and daily study hours. A graph-based similarity search, powered by GraphRAG, recommends practice questions tagged by difficulty and module. Delivered via a Streamlit interface with interactive elements like countdown timers and topic priority visualizations, the system ensures scalability, privacy through local data processing, and adaptability to other SLIIT modules, ultimately enhancing student engagement, exam performance, and long-term knowledge retention.

2.0 OBJECTIVES

2.1 Main Objectives

Develop a scalable NLP-driven framework for the DMS module at SLIIT to semantically classify exam questions and lecture materials into hierarchical syllabus modules, analyse a decade of historical exam trends, and generate time-aware, personalized study plans based on exam dates and daily study hours, with practice question recommendations to optimize exam preparation efficiency and enhance curriculum alignment for improved student engagement and academic performance.

2.2 Specific Objectives

1 **Data Acquisition & Preprocessing**

- Collect and preprocess a decade of SLIIT DMS MCQ exam papers and lecture slides, applying cleaning, tokenization, and OCR error correction to ensure high-quality, usable data for analysis.

2 **Syllabus Module Classification**

- Classify exam questions and lecture materials into DMS syllabus modules (e.g., Normalization, SQL Queries) using Sentence-BERT and fine-tuned classifiers, producing accurately mapped content with module tags.

3 **Historical Exam Trend Analysis**

- Analyse a decade of SLIIT DMS exam data to identify high-priority topics based on frequency and syllabus coverage, generating a ranked topic list aligned with exam patterns.

4 **Personalized Study Plan Generation**

- Develop a scheduling algorithm that uses exam dates and daily study hours to create time-aware, personalized study plans, prioritizing high-impact topics for efficient revision.

5 **Practice Question Recommendation**

- Implement an embedding-based similarity search using GraphRAG to recommend the top five relevant past exam questions per topic, tagged by difficulty and syllabus module.

6 **User Interface Development**

- Design a Streamlit-based interface featuring countdown timers, topic priority visualizations, progress trackers, and search/filter options to enhance usability and encourage high student engagement.

7 System Validation and Robustness

- Validate the system's classification accuracy, study plan effectiveness through user feedback, and robustness across SLIIT datasets, ensuring alignment with the DMS syllabus and reliable performance.

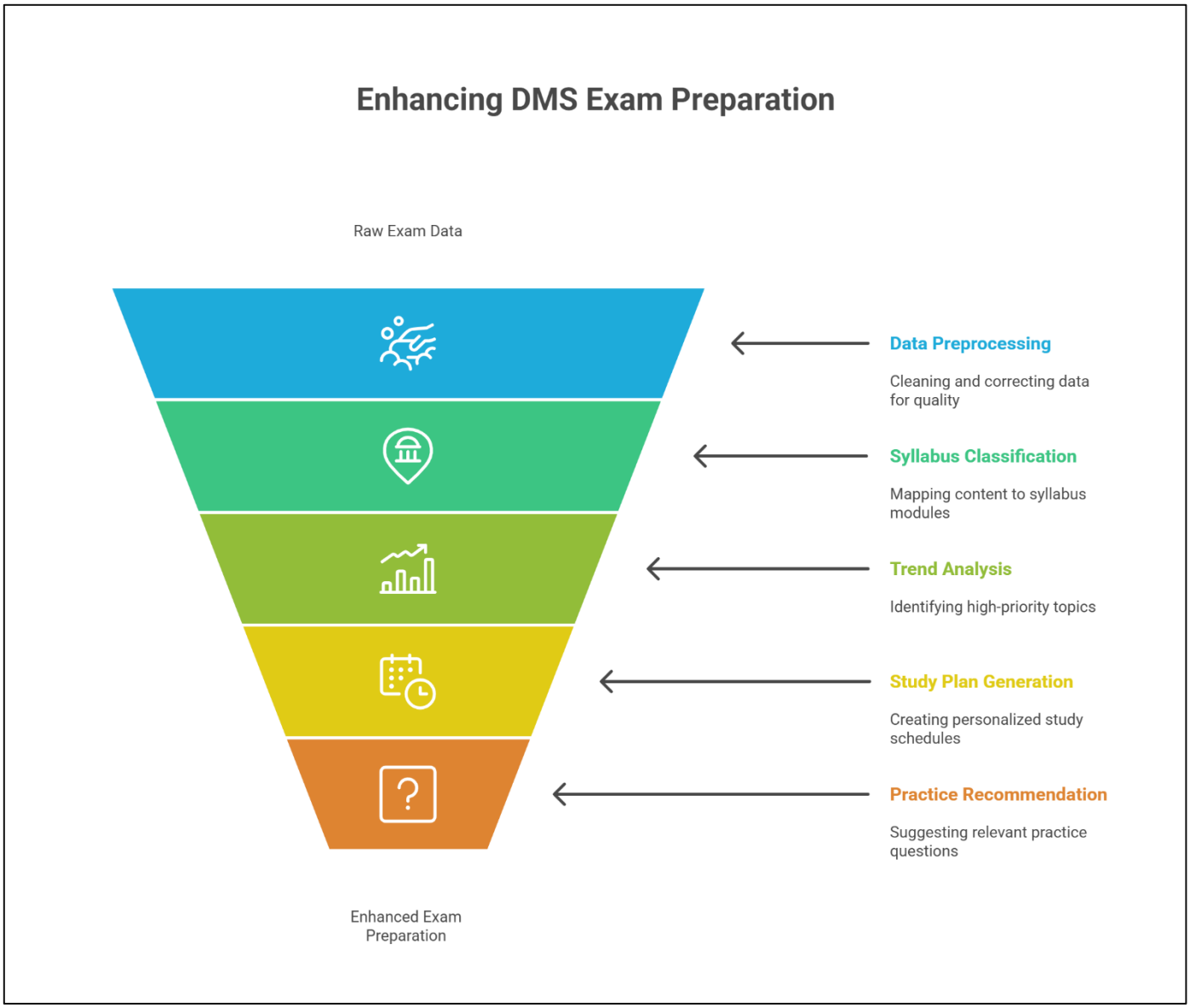

Figure 2: Study Plan System

3.0 METHODOLOGY

3.1 System description

This project develops an NLP-driven system tailored for the Database Management Systems (DMS) module at the Sri Lanka Institute of Information Technology (SLIIT) to optimize exam preparation by aligning lecture content with assessment patterns, addressing the main objective of enhancing curriculum alignment and student performance. The system collects a decade of DMS MCQ exam papers and lecture slides from SLIIT's academic repositories, preprocesses them to create a clean dataset, and employs Sentence-BERT for semantic classification to map content into hierarchical syllabus modules (e.g., Normalization, SQL Queries). Frequency analysis identifies high-priority topics based on historical exam trends, while a scheduling algorithm generates time-aware, personalized study plans using student inputs (exam dates, daily study hours). An embedding-based similarity search recommends relevant practice questions, tagged by difficulty and module. A Streamlit-based interface delivers interactive features like countdown timers, topic priority visualizations, and progress trackers. Local processing ensures privacy compliance, and the modular design supports scalability and adaptation to other SLIIT modules, achieving efficient exam preparation and curriculum alignment.

3.2 Key Components

1. **Data Collection & Preprocessing:**
    - Sources: Collect a decade of SLIIT DMS MCQ exam papers (PDF/scans) and lecture slides (PDF/PPT/DOCX) from SLIIT's academic repositories with institutional approval.
    - Tools: Use Tesseract OCR for text extraction, NLTK for tokenization, and custom Python scripts for noise removal and error correction (e.g., Levenshtein distance-based spell-checking, BERT-based contextual correction).
    - Process: Normalize English text (lowercase, remove special characters), tag entities (e.g., Normalization) using Named Entity Recognition (NER), and store in an SQLite database.
    - Output: Cleaned, tokenized dataset suitable for analysis.
2. **Syllabus Module Classification:**
    - Labeling: Manually label a subset of data with DMS syllabus module tags based on SLIIT's curriculum.
    - Classifier: Train a fine-tuned BERT classifier for multi-class classification of content into modules (e.g., Normalization, SQL Queries).

- Validation: Use 5-fold cross-validation with dropout to ensure robust performance and prevent overfitting.
- Output: Classified dataset with module tags.

3. **Topic Importance Analysis:**
    - Method: Apply TF-IDF to extract key terms and compute question frequency across a decade of DMS exams using Pandas.
    - Weighting: Assign importance scores based on question recurrence and syllabus coverage.
    - Output: Ranked topic list with bar chart visualizations (using Matplotlib/Seaborn).

4. **Personalized Study Plan Generation:**
    - Inputs: Collect exam dates and daily study hours via a Streamlit form.
    - Algorithm: Implement a rule-based scheduling algorithm to allocate study time based on topic importance and days remaining.
    - Output: Phased study plans (e.g., "Day 1: 3h Normalization, 1h SQL Joins") in PDF and dashboard formats.

5. **Practice Question Recommendation:**
    - Method: Use cosine similarity on Sentence-BERT embeddings integrated with graph structures to identify similar past questions.
    - Indexing: GraphRAG for efficient graph-based retrieval (k=5), leveraging knowledge graphs to model relationships between topics, modules, and questions.
    - Tagging: Assign difficulty levels based on historical marks and Bloom's Taxonomy.
    - Output: Top five questions per topic, tagged with module and difficulty.

6. **User Interface Development:**
    - Framework: Develop a Streamlit interface hosted locally or on AWS.
    - Features: Implement input forms, countdown timers, topic priority graphs, progress trackers, and search/filter options.
    - Output: Responsive dashboard with PDF/CSV export functionality.

7. **Validation and Scalability:**
    - Validation: Test classification accuracy, study plan effectiveness via user feedback, and system robustness using SLIIT datasets.
    - Scalability: Optimize for large datasets using Dask for parallel processing.
    - Adaptability: Design modular code to enable retraining for other SLIIT modules.
    - Output: Validated system aligned with DMS syllabus and user needs.

3.3 System Workflow

- **Data Input:**
    - Ingest SLIIT DMS exam papers and lecture slides (PDF/PPT/DOCX) via file upload.
    - Apply Tesseract OCR for text extraction, NLTK for tokenization, and Named Entity Recognition (NER) to tag topics (Normalization).
- **Content Processing:**
    - Generate Sentence-BERT embeddings for questions and slides.
    - Classify content into DMS syllabus modules using fine-tuned BERT
    - Analyze topic frequency using TF-IDF and Pandas.
- **Personalization Layer:**
    - Collect exam date and daily study hours via Streamlit UI.
    - Generate time-aware study plans using the scheduling algorithm.
    - Retrieve top-5 practice questions per topic via GraphRAG-based similarity search.
- **Output Delivery:**
    - Display study plans, question recommendations, and visualizations (priority graphs) on a Streamlit dashboard.

3.4 Tools & Technology

- **Programming:** Python (primary), JavaScript (front end).
- **NLP & ML:** PyTorch, TensorFlow, spaCy, scikit-learn, HuggingFace Transformers, GraphRAG
- **OCR:** Tesseract OCR
- **Databases:** MongoDB**,** Elasticsearch
- **Web Front End:** React.js; Tailwind CSS, Streamlit.
- **Design/Diagrams:** Figma; draw.io.
- **IDE:** VS Code, Google Colab, Jupyter Notebook, Kaggle Platform
- **VCS:** GitHub
- **Collaboration:** Microsoft Teams, Slack.
- **Testing:** pytest (Python), Postman (APIs), Jest (JS).
- **Deployment:** Docker, Kubernetes, Jenkins (CI/CD)
- **Project Management:** Jira

Overall System Diagram

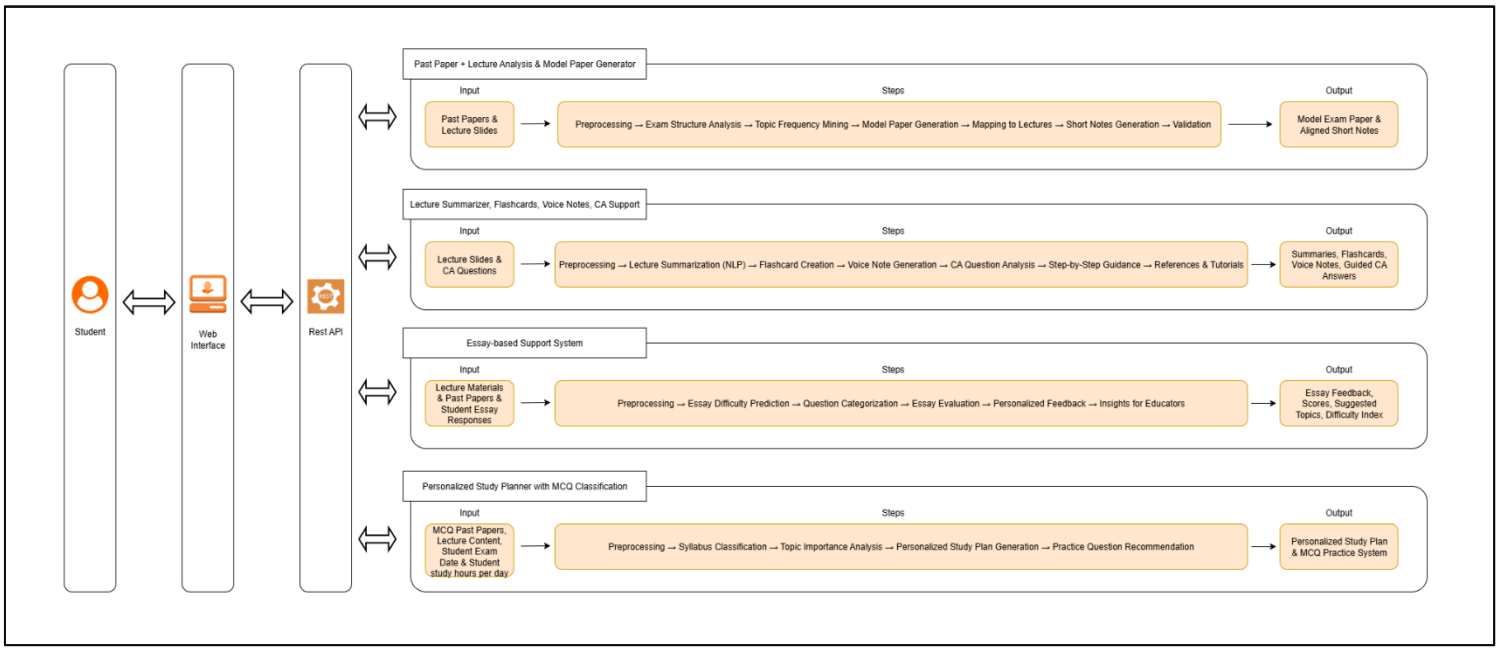

Figure 3: System Diagram

Component Flow Diagram

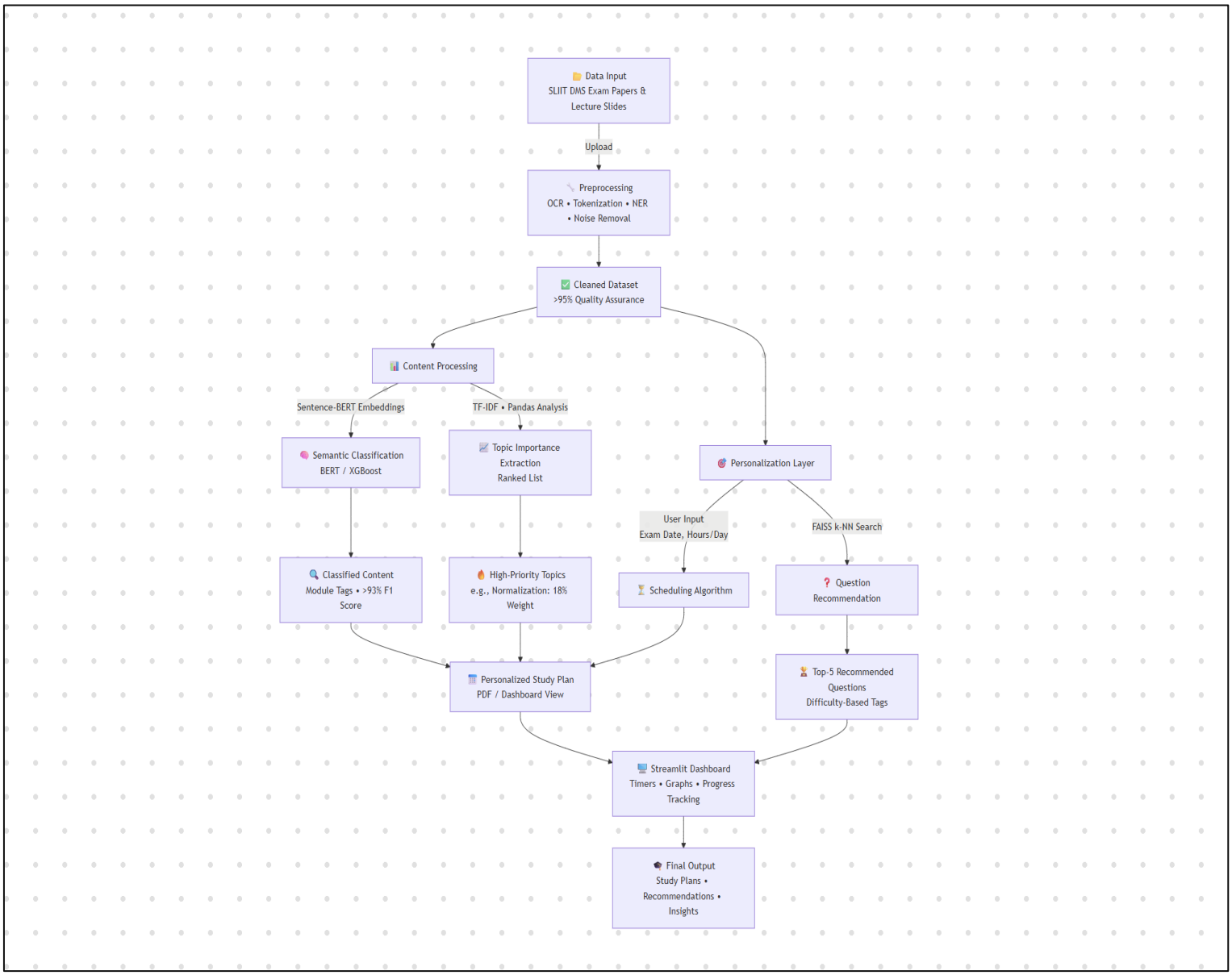

Figure 4: Component Diagram

Figure 5: Work breakdown Structure

3.4 PROJECT REQUIREMENTS

3.4.1 Functional requirements

1. **Data Input and Processing**

- Support multiple file formats (PDF, PPT, DOCX) for exam papers and lecture slides.
- Use Tesseract OCR to extract text from scanned documents.
- Perform preprocessing steps including cleaning, tokenization, noise removal, and error correction (Levenshtein distance, BERT-based correction).
- Recognize syllabus-related entities (e.g.,Normalization, SQL Queries) using Named Entity Recognition (NER).

2. Content Analysis and Classification

- Classify exam questions and lecture materials into DMS syllabus modules.
- Utilizes semantic embeddings to facilitate similarity-based comparisons.
- Provide hierarchical mapping of content to syllabus structures.
- Analyze historical exam data to identify recurring and high-priority topics.

3. Personalization and Recommendation

- Generates personalized study plans tailored to exam dates and available study hours.
- Allocates study time according to the importance and weight of topics.
- Recommends practice questions through graph-based semantic similarity matching.
- Tags questions with metadata such as syllabus module and difficulty level.

4. User Interface and Interaction

- Offers an interactive dashboard with visualizations including countdown timers, topic graphs, and progress trackers.
- Allows students to search, filter, and customize study schedules.
- Provides export options for study plans and question sets in formats

5. Error Handling

- Provides user notifications for incomplete, corrupted, or unreadable files.
- Includes fallback mechanisms for handling noisy or low-quality input data.

3.4.2 Non-functional requirements

- **Performance**
  - Capable of processing uploaded documents efficiently without noticeable delays.
  - Generates study plans and recommendations with minimal latency during user interaction.
- **Scalability**
  - Designed to handle large volumes of academic documents and questions.
  - Structured to extend easily to other academic modules beyond DMS.
- **Usability**
  - Ensures a clean, intuitive, and user-friendly interface with straightforward navigation.
  - Encourages engagement through progress tracking, reminders, and interactive features.
- **Security and Privacy**
  - Protects sensitive data through encryption in both storage and transmission.
  - Processes academic information within secure and compliant environments.
- **Reliability**
  - Maintains consistent availability with minimal downtime.
  - Recovers gracefully from unexpected errors or failures.
- **Maintainability**
  - Built on modular, well-documented components to simplify debugging and updates.
  - Supports easy retraining or adaptation of models for new syllabus content.
- **Interoperability**
  - Ensure compatibility with standard file formats (PDF, PPT, DOCX).
  - Facilitates integration with third-party analytics and reporting tools.

3.5 Expected Test Cases

Test Case ID	Input	Action	Expected Output
TC-01	Upload past exam papers (PDF/DOCX)	Preprocess and extract text	Cleaned, tokenized exam question dataset generated
TC-02	Upload lecture slides (PPT/PDF)	Extract lecture content	Preprocessed lecture dataset ready for mapping
TC-03	Input syllabus module (e.g., "Normalization")	Run Sentence-BERT classification	Questions and slides mapped correctly to the selected module
TC-04	Past papers contain repeated topics	Perform frequency analysis	High-frequency topics identified and ranked for prioritization
TC-05	Student enters exam date and daily study hours	Trigger scheduling algorithm	Personalized, time-aware study plan generated
TC-06	Student requests practice questions	System runs GraphRAG similarity search	Relevant practice questions displayed, tagged by difficulty
TC-07	User opens Streamlit interface	Load dashboard	Countdown timer, topic priority chart, and progress tracker visible
TC-08	Multiple students use system with different inputs	Validate modularity	Each student receives unique, personalized study plan
TC-09	Upload large dataset (10 years of papers & slides)	Run system processing	System handles load without crash (scalability test)
TC-10	User provides no input or invalid data	System validation	Error message displayed (e.g., “Please upload valid file”)

3.6 Timeline

The proposed timeline for the project is as follows.

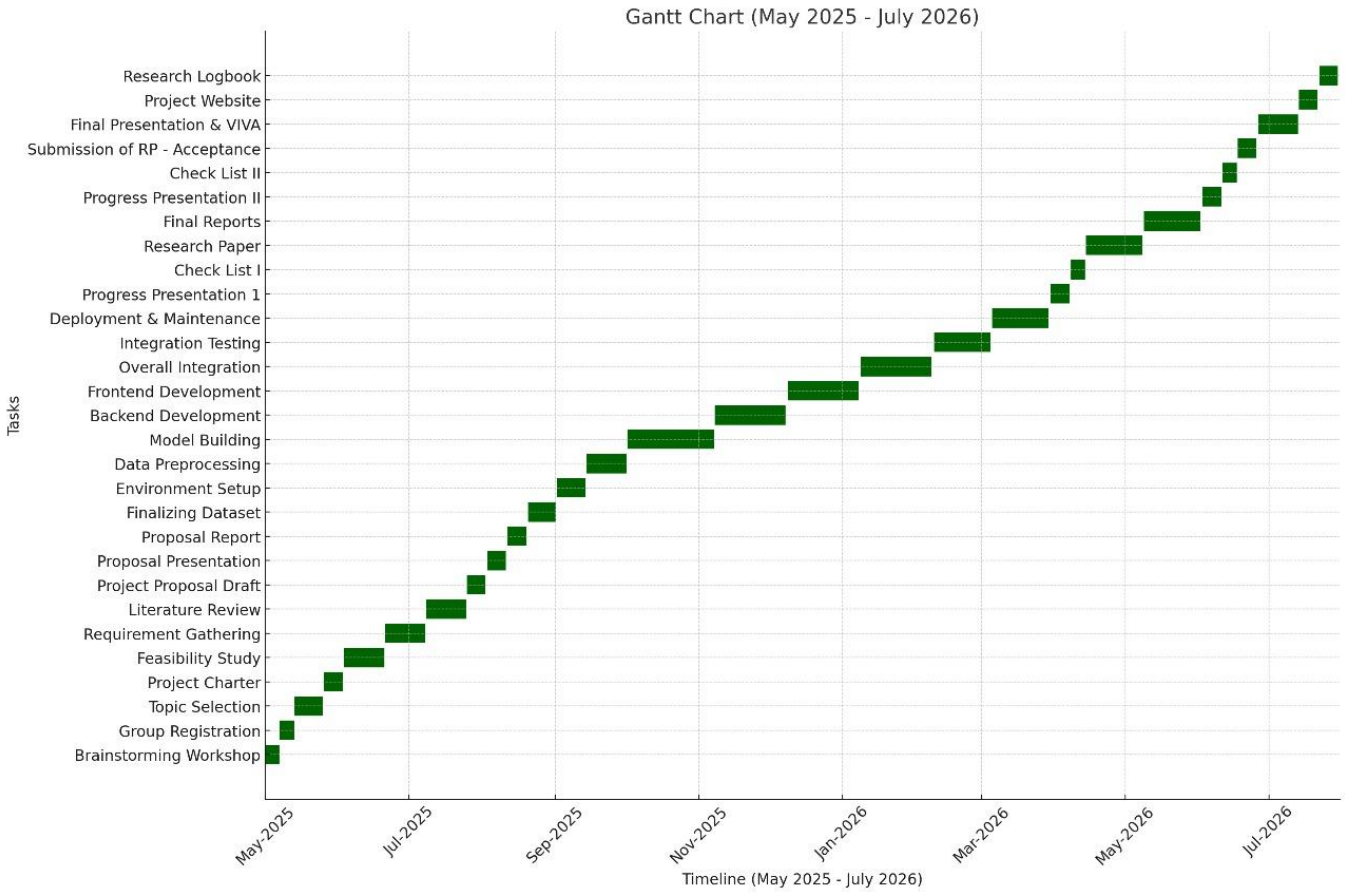

Figure 6:Gantt Chart for Project Schedule Management

3.7 Risk Management Plan

To identify, assess, and mitigate risks throughout the development and deployment of the NLP-driven system for semantic classification and personalized study plan generation, ensuring successful project completion within scope, time, and budget.

Risk	Description	Probability	Impact	Mitigation Strategy	Contingency Plan
Technical	Inaccurate classification due to noisy OCR data (e.g., exam paper scans).	Medium	High	Use Tesseract OCR with Levenshtein-based correction; validate with 5-fold cross-validation in Google Colab/Kaggle.	Manually preprocess a subset of data; retrain models with synthetic datasets.
Operational	Delay in accessing SLIIT exam papers/slides	High	High	Secure approvals in Month 1 ; coordinate via Microsoft Teams with supervisor.	Use publicly available DMS datasets or synthetic data to proceed with development.
Data-Related	Privacy breaches in handling exam data.	Low	High	Process data locally with SQLite; anonymize datasets; comply with SLIIT/GDPR policies.	conduct security audit if breach detected.
External	Syllabus changes affecting model relevance.	Low	Medium	Monitor updates via supervisor meetings; design modular system for retraining in Visual Studio Code.	Retrain models with updated syllabus data; adjust study plan algorithms.
External	Budget overruns for cloud resources (e.g., Colab/Kaggle).	Medium	Medium	Track costs monthly via Jira; prioritize free tiers of Colab/Kaggle and open-source tools.	Shift to local processing with Visual Studio Code; reduce model complexity to minimize resource use.

4.0 COMMERCIALIZATION

The proposed NLP-driven educational framework for semantic classification and personalized study plan generation holds strong commercialization potential within the higher education and EdTech sectors. With the rising demand for AI-powered, personalized learning tools that address curriculum alignment and improve exam preparation efficiency, the system enters a promising market. Its strengths lie in privacy-compliant local data processing, a scalable architecture adaptable to multiple academic modules, and the integration of advanced NLP techniques for semantic content mapping and personalized scheduling. Target customers include universities, colleges, online platforms, and private tutoring services seeking to boost student engagement and outcomes. The commercialization strategy could take the form of licensing the software to institutions or offering it as a subscription-based SaaS platform with advanced analytics features. Pilot deployments at institutions like the Sri Lanka Institute of Information Technology (SLIIT) would validate the concept and provide valuable user feedback for refinement, while partnerships with EdTech companies could support wider distribution. Safeguarding intellectual property through patents or copyrights and establishing strong technical support and training infrastructure will be vital for success. Ultimately, by turning data-driven insights into actionable, personalized learning experiences, the solution is well-positioned to address the evolving needs of education providers and students alike.

5.0 BUDGET & BUDGET JUSTIFICATION

Since the outcome of the proposed model is a software-based solution, there are no hardware components connected to the implementation. The primary source of the cost will be the subscription fees to the cloud provider for the computing power of the virtual machines. However, there will be some other costs expected to as be given in the table below.

Type	**Cost**
Internet use and web hosting	10 000 LKR
Training Cost	10 000 LKR
Publication Cost	50 000 LKR
Stationery	1 000 LKR
Additional	10 000 LKR
TOTAL	***81 000 LKR***

6.0 SUMMARY

This project proposes an AI-powered, NLP-driven framework to enhance exam preparation and curriculum alignment for the Database Management Systems (DMS) module at the Sri Lanka Institute of Information Technology (SLIIT). It addresses key challenges faced by DMS students, such as aligning unstructured lecture materials with syllabus modules and prioritizing high-impact topics for efficient study. By analyzing a decade of SLIIT DMS exam papers and lecture content, the system uses Sentence-BERT for semantic classification to map materials to syllabus modules like Normalization and SQL Queries. Frequency analysis identifies recurring topics, while a scheduling algorithm generates personalized, time-aware study plans based on student inputs, including exam dates and daily study hours. A graph-based similarity search via GraphRAG recommends relevant practice questions, tagged by difficulty and module, to support targeted revision.

Delivered through an interactive Streamlit interface, the system features countdown timers, topic priority visualizations, and progress trackers to boost student engagement and usability. Local data processing ensures privacy compliance, and the modular design enables scalability to other SLIIT modules. Rigorous validation confirms high classification accuracy, user satisfaction, and robust performance across diverse datasets. By optimizing exam preparation, enhancing curriculum alignment, and improving academic performance, this framework offers a scalable, privacy-compliant solution that promotes long-term knowledge retention and supports institutional teaching quality, significantly benefiting DMS students and potentially other academic programs at SLIIT.

7.0 References

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